

Exp. 2	DEVELOP A SIMPLE APPLICATION TO UNDERSTAND THE CONCEPT OF PAAS USING GAE/AMAZON ELASTIC BEANSTALK/IBM BLUE MIX/GCC AND LAUNCH IT
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AIM

To develop and deploy a simple web application using a Platform as a Service (PaaS) cloud environment such as Google App Engine, Amazon Elastic Beanstalk, or IBM Cloud.

OBJECTIVE

- To understand the concept of Public Cloud Computing.
- To learn the Platform as a Service (PaaS) model.
- To deploy and manage applications in a cloud environment.
- To understand cloud-based application hosting and scaling.

SOFTWARE REQUIREMENTS

- Google Cloud Platform (GCP) / AWS Account
- Google App Engine or Amazon Elastic Beanstalk
- Python/Java/Node.js Runtime
- Web Browser
- Internet Connection
- Text Editor or IDE

THEORY

Public Cloud is a cloud deployment model in which computing resources such as servers, storage, databases, and networking are owned and managed by a third-party cloud provider and delivered over the internet. Users can access these resources on demand and pay only for what they use. Platform as a Service (PaaS) is a cloud service model that provides developers with a complete environment for developing, testing, deploying, and managing applications. The cloud provider manages the underlying infrastructure, operating systems, middleware, and runtime environments, allowing developers to focus only on application development. Examples of PaaS platforms include Google App Engine, Amazon Elastic Beanstalk, Microsoft Azure App Service, and IBM Cloud.

ADVANTAGES OF PAAS

- Faster application development and deployment.
- Reduced infrastructure management.

- Automatic scaling and load balancing.

ARCHITECTURE

PaaS Architecture:



ALGORITHM

1. Create an account on the cloud platform.
2. Develop a simple web application.
3. Configure the application deployment settings.
4. Install and configure Cloud SDK.
5. Create a cloud project.
6. Deploy the application to the cloud platform.
7. Verify application deployment.
8. Access the application through the generated URL.

FLOWCHART



Access Application URL

↓

Stop

PROCEDURE

STEP 1: CREATE A CLOUD ACCOUNT

1. Create an account on Google Cloud Platform or AWS.
2. Sign in to the cloud console.
3. Enable billing if required.

STEP 2: CREATE A SIMPLE APPLICATION

Create a sample web application.

Example (Python Flask):

```
from flask import Flask

app = Flask(__name__)

@app.route('/')
def home():
    return "Hello Cloud Computing"

if __name__ == '__main__':
    app.run()
```

STEP 3: CREATE APPLICATION CONFIGURATION FILE

Create a configuration file.

Example (app.yaml):

```
runtime: python39

handlers:
- url: /*.*
  script: auto
```

STEP 4: INSTALL CLOUD SDK

Install Google Cloud SDK or AWS CLI.

Verify installation:

```
gcloud version
```

STEP 5: LOGIN TO CLOUD PLATFORM

```
gcloud auth login
```

STEP 6: CREATE CLOUD PROJECT

Create a new project from the Google Cloud Console.

Project Name: cloud-lab-project

Alternatively, a project may be created using the Google Cloud SDK:

```
gcloud projects create cloud-lab-project
```

STEP 7: SET ACTIVE PROJECT

Configure the newly created project as the active project:

```
gcloud config set project cloud-lab-project
```

Verify the configuration:

```
gcloud config list
```

STEP 8: DEPLOY APPLICATION

Deploy the application:

```
gcloud app deploy
```

STEP 9: LAUNCH APPLICATION

```
gcloud app browse
```

STEP 10: VERIFY DEPLOYMENT

1. Open the generated URL.
2. Verify that the application loads successfully.
3. Test application functionality.

COMMANDS USED

```
gcloud auth login
gcloud projects create cloud-lab-project
gcloud config set project cloud-lab-project
gcloud app create
```

```
gcloud app deploy  
gcloud app browse
```

OBSERVATIONS

Parameter	Status
Cloud Account Creation	Successful
Project Creation	Successful
Application Development	Successful
Application Deployment	Successful
URL Generation	Successful
Application Access	Verified

EXPECTED OUTPUT

- Cloud project created successfully.
- Application deployed on the cloud platform.
- Public URL generated.
- Application accessible through a web browser.
- Successful execution of application services.

RESULT

A simple web application was successfully deployed on a Public Cloud Platform using the PaaS model. The application was accessible through a web browser and managed by the cloud provider.